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MacroSight™ — V1_FULL CONTRACT_METHODS

ROLE DEFINITIONS

Role	Scope
SUPER_ADMIN	Platform, security, approvals, overrides
ECONOMIC_ANALYST	Forecasting, scenarios, interpretation
DATA_SCIENTIST	Model lifecycle, metrics, explainability, train model
AUDITOR	Independent audit & compliance
EXECUTIVE_VIEWER	Read-only insights
DATA_FEEDER (Company Staff)	Data ingestion only

1. AUTHENTICATION & ACCESS API

Access: Public → All Roles

Method	Endpoint
POST	/api/v1/auth/login/
POST	/api/v1/auth/refresh/
POST	/api/v1/auth/logout/
GET	/api/v1/auth/me/
POST	/api/v1/auth/change-password/
POST	/api/v1/auth/mfa/verify/

Admin Extensions

Method	Endpoint
GET	/api/v1/admin/auth/login-history/
POST	/api/v1/admin/auth/force-logout/{user_id}/

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2. USERS & ROLES API

Access: SUPER_ADMIN

Users

Method	Endpoint
GET	/api/v1/admin/users/
POST	/api/v1/admin/users/
GET	/api/v1/admin/users/{id}/
PUT	/api/v1/admin/users/{id}/
DELETE	/api/v1/admin/users/{id}/
PUT	/api/v1/admin/users/{id}/disable/
PUT	/api/v1/admin/users/{id}/enable/
PUT	/api/v1/admin/users/{id}/reset-credentials/

Roles & Permissions

Method	Endpoint
GET	/api/v1/admin/roles/
POST	/api/v1/admin/roles/
PUT	/api/v1/admin/roles/{id}/
DELETE	/api/v1/admin/roles/{id}/
PUT	/api/v1/admin/users/{id}/role/
GET	/api/v1/admin/permissions/

3. SALES DATA API

Data Feeder

Method	Endpoint
POST	/api/v1/sales/upload/
GET	/api/v1/sales/upload/status/

Analyst

Method	Endpoint
GET	/api/v1/sales/
GET	/api/v1/sales/{id}/
GET	/api/v1/sales/export/

Admin Controls

Method	Endpoint
DELETE	/api/v1/sales/{id}/
PUT	/api/v1/admin/sales/{id}/lock/
GET	/api/v1/admin/sales/ingestion-logs/
GET	/api/v1/admin/sales/data-quality/

4. PRODUCT API

Role	Endpoint
Analyst	GET /api/v1/products/
SuperAdmin	POST /api/v1/products/
SuperAdmin	PUT /api/v1/products/{id}/
SuperAdmin	DELETE /api/v1/products/{id}/
Auditor	GET /api/v1/products/history/

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5. EXTERNAL INDICATORS API

Analyst

Method	Endpoint
GET	/api/v1/indicators/
POST	/api/v1/indicators/
PUT	/api/v1/indicators/{id}/

Admin

Method	Endpoint

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DELETE	/api/v1/indicators/{id}/
POST	/api/v1/admin/indicators/sync/
PUT	/api/v1/admin/indicators/source/
GET	/api/v1/admin/indicators/quality-report/
GET	/api/v1/admin/indicators/ingestion-log/

MANUAL MACRO INDICATORS API

Purpose: Allow analysts to **physically update structural macro parameters**.

Access:

- ECONOMIC_ANALYST (create/update)
- SUPER_ADMIN (approve)
- AUDITOR (read)

Indicator Parameter Registry

Method	Endpoint
GET	/api/v1/macro/parameters/
POST	/api/v1/macro/parameters/
PUT	/api/v1/macro/parameters/{id}/
DELETE	/api/v1/macro/parameters/{id}/

Financial-Year & Effective-Date Control

Method	Endpoint
POST	/api/v1/macro/parameters/{id}/version/
PUT	/api/v1/macro/parameters/{vid}/activate/

6

. POLICY & LEGISLATION API	
Role	**Endpoint**

V1 CONTRACTS

SuperAdmin	POST /api/v1/policies/
Analyst	POST /api/v1/policies/{id}/impact/
Executive	GET /api/v1/policies/
SuperAdmin	PUT /api/v1/policies/{id}/
SuperAdmin	DELETE /api/v1/policies/{id}/
Auditor	GET /api/v1/policies/history/

7. FORECAST MODEL API (FULL LIFECYCLE)

Data Scientist

Method	Endpoint
POST	/api/v1/models/register/
PUT	/api/v1/models/{id}/
GET	/api/v1/models/{id}/metrics/
GET	/api/v1/models/{id}/explainability/

Governance (Admin)

Method	Endpoint
POST	/api/v1/admin/models/{id}/promote/
POST	/api/v1/admin/models/{id}/rollback/
DELETE	/api/v1/models/{id}/
GET	/api/v1/admin/models/risk-report/

8. FORECAST RESULTS API

Role	Endpoint
Analyst	POST /api/v1/forecasts/generate/
Analyst	GET /api/v1/forecasts/
Executive	GET /api/v1/forecasts/{id}/correctness-summary/

Executive	GET /api/v1/forecasts/{id}/
Analyst	GET /api/v1/forecasts/{id}/explain/
Admin	PUT /api/v1/admin/forecasts/{id}/invalidate/

9. SCENARIO SIMULATION API

Role	Endpoint
Analyst	POST /api/v1/scenarios/run/
Analyst	GET /api/v1/scenarios/
Executive	GET /api/v1/scenarios/{id}/compare/
Analyst	DELETE /api/v1/scenarios/{id}/
Admin	GET /api/v1/admin/scenarios/audit/

10. GOVERNANCE & AUDIT API

Access: SUPER_ADMIN, AUDITOR
READ-ONLY — NO ADDITIONS ALLOWED

Question	Endpoint
Forecast provenance	/api/v1/audit/forecast-lineage/{id}/
Model reliability	/api/v1/audit/model-runs/
Data integrity	/api/v1/audit/data-snapshot/
Indicator influence	/api/v1/audit/indicator-impact/{id}/
Policy effect	/api/v1/audit/policy-impact/{id}/

11. SYSTEM & ADMIN OPS API

Category	Endpoint
Health	GET /api/v1/system/health/
Config	GET /api/v1/system/config/
Config	PUT /api/v1/system/config/

Retraining	POST /api/v1/system/retrain/
Backup	POST /api/v1/system/backup/
Jobs	GET /api/v1/admin/jobs/
Jobs	POST /api/v1/admin/jobs/{id}/pause/
Jobs	POST /api/v1/admin/jobs/{id}/resume/
Jobs	GET /api/v1/admin/jobs/history/
Usage	GET /api/v1/admin/system/usage/

12. SECTOR REGISTRY API

Defines economic sectors used across forecasting, policy impact, and dashboards.

Access: SUPER_ADMIN, ECONOMIC_ANALYST (read)

Sector Management

Method	Endpoint
GET	/api/v1/sectors/
POST	/api/v1/sectors/
GET	/api/v1/sectors/{id}/
PUT	/api/v1/sectors/{id}/
DELETE	/api/v1/sectors/{id}/

Sector Metadata

Method	Endpoint
GET	/api/v1/sectors/{id}/indicators/
GET	/api/v1/sectors/{id}/policies/
GET	/api/v1/sectors/{id}/cost-drivers/

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13. POLICY PARAMETERS (VERSIONED) API

Used for **annual financial year updates** (taxes, subsidies, duties).

Access:

- SUPER_ADMIN (write)

VI CONTRACTS

- ECONOMIC_ANALYST (propose & simulate)
- AUDITOR (read-only)

Policy Parameter Registry

Method	Endpoint
GET	/api/v1/policies/parameters/
POST	/api/v1/policies/parameters/
PUT	/api/v1/policies/parameters/{id}/
DELETE	/api/v1/policies/parameters/{id}/

Financial Year Versioning

Method	Endpoint
GET	/api/v1/policies/financial-years/
POST	/api/v1/policies/financial-years/
GET	/api/v1/policies/financial-years/{fy}/
PUT	/api/v1/policies/financial-years/{fy}/activate/

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14. SECTOR–POLICY MAPPING API

Defines **which policies affect which sectors**.

Access: SUPER_ADMIN

Method	Endpoint
POST	/api/v1/policies/sector-map/
GET	/api/v1/policies/sector-map/
DELETE	/api/v1/policies/sector-map/{id}/

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15. INPUT COST INTELLIGENCE API

Captures **cost drivers that affect demand & profitability**.

Input Cost Registry

Access: ECONOMIC_ANALYST, DATA_FEEDER (submit), SUPER_ADMIN (override)

Method	Endpoint
GET	/api/v1/costs/
POST	/api/v1/costs/
PUT	/api/v1/costs/{id}/

DELETE	/api/v1/costs/{id}/

Sector-Specific Cost Profiles

Method	Endpoint
GET	/api/v1/sectors/{id}/cost-profile/
POST	/api/v1/sectors/{id}/cost-profile/
PUT	/api/v1/sectors/{id}/cost-profile/{cid}/

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16. AGRICULTURAL INPUTS API

Specialized inputs for **fertilizers, seeds, pests, fuel**.

Access: DATA_FEEDER (submit), ANALYST (analyze)

Method	Endpoint
POST	/api/v1/agri/inputs/fertilizers/
POST	/api/v1/agri/inputs/seeds/
POST	/api/v1/agri/inputs/pesticides/
POST	/api/v1/agri/inputs/fuel/
GET	/api/v1/agri/inputs/
GET	/api/v1/agri/inputs/summary/

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17. SECTOR DEMAND FORECAST API

Forecasts demand **by sector, policy version, and cost snapshot**.

Access: ECONOMIC_ANALYST, EXECUTIVE_VIEWER (read)

Method	Endpoint
POST	/api/v1/forecasts/sector-demand/run/
GET	/api/v1/forecasts/sector-demand/
GET	/api/v1/forecasts/sector-demand/{id}/

---

18. SECTOR PROFITABILITY FORECAST API

Evaluates **margin sensitivity to policy & input costs**.

Access: ECONOMIC_ANALYST

Method	Endpoint

V1 CONTRACTS

POST	/api/v1/forecasts/profitability/run/
GET	/api/v1/forecasts/profitability/
GET	/api/v1/forecasts/profitability/{id}/

---

19. POLICY IMPACT SIMULATION API

"What if VAT increases next FY?"

Access: ECONOMIC_ANALYST

Method	Endpoint
POST	/api/v1/policies/simulate/
GET	/api/v1/policies/simulations/
GET	/api/v1/policies/simulations/{id}/

---

20. FORECAST COMPARISON & DELTA API

Compare **before vs after policy or cost change**.

Access: ANALYST, EXECUTIVE

Method	Endpoint
GET	/api/v1/forecasts/compare/
GET	/api/v1/forecasts/{id}/delta/

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21. GOVERNMENT NOTICE & ADVISORY API

Captures **policy announcements & regulatory notices**.

Access: ECONOMIC_ANALYST (create), EXECUTIVE (read)

Method	Endpoint
POST	/api/v1/notices/government/
GET	/api/v1/notices/government/
GET	/api/v1/notices/government/{id}/

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22. POLICY CHANGE APPROVAL API

Prevents unauthorized policy activation.

Access: SUPER_ADMIN

V1 CONTRACTS

Method	Endpoint
GET	/api/v1/admin/policies/pending/
POST	/api/v1/admin/policies/{id}/approve/
POST	/api/v1/admin/policies/{id}/reject/

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23. POLICY & COST AUDIT EXTENSIONS

(READ-ONLY)

Question	Endpoint
Policy version used	/api/v1/audit/policy-version/{forecast_id}/
Sector sensitivity	/api/v1/audit/sector-impact/{sector_id}/
Cost change trace	/api/v1/audit/cost-history/{id}/
Profitability drivers	/api/v1/audit/profitability-breakdown/{id}/

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24. DASHBOARD DATA AGGREGATION API

Feeds role-specific dashboards efficiently.

Access: Role-bound

Role	Endpoint
Executive	/api/v1/dashboard/executive/
Analyst	/api/v1/dashboard/analyst/
Auditor	/api/v1/dashboard/audit/
Data Feeder	/api/v1/dashboard/data-feeder/

V1 CONTRACTS

TRAINING APIS FOR DATA SCIENTIST

TRAINING DATA PREPARATION API

1. Training Dataset Builder

Builds **versioned datasets** from approved sources.

Method	Endpoint

POST	/api/v1/ml/datasets/build/
GET	/api/v1/ml/datasets/
GET	/api/v1/ml/datasets/{id}/
DELETE	/api/v1/ml/datasets/{id}/

Inputs

- Sales data snapshot
- Indicator version
- Policy version
- Sector scope
- Date range

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2. Feature Engineering API

Transforms raw data → ML-ready features.

Method	Endpoint
POST	/api/v1/ml/features/generate/
GET	/api/v1/ml/features/
GET	/api/v1/ml/features/{id}/
PUT	/api/v1/ml/features/{id}/
DELETE	/api/v1/ml/features/{id}/

Examples:

- Lag features
- Rolling averages
- Policy interaction terms
- Cost sensitivity ratios

B. MODEL TRAINING API

3. Training Job API

Runs training asynchronously.

Method	Endpoint
POST	/api/v1/ml/train/
GET	/api/v1/ml/train/jobs/
GET	/api/v1/ml/train/jobs/{id}/
POST	/api/v1/ml/train/jobs/{id}/cancel/

V1 CONTRACTS

Training configs

- Algorithm (XGBoost, LSTM, TFT)
- Hyperparameters
- Dataset ID
- Feature set ID

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4. Hyperparameter Search API

(Optional but powerful)

Method	Endpoint
POST	/api/v1/ml/hpo/run/
GET	/api/v1/ml/hpo/runs/
GET	/api/v1/ml/hpo/runs/{id}/

C. MODEL EVALUATION API

5. Validation & Metrics API

Method	Endpoint
GET	/api/v1/ml/models/{id}/metrics/
GET	/api/v1/ml/models/{id}/cross-validation/
GET	/api/v1/ml/models/{id}/backtest/

Metrics:

- RMSE
- MAPE
- MAE
- Coverage (for confidence intervals)

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6. Model Comparison API

Method	Endpoint
GET	/api/v1/ml/models/compare/

D. MODEL EXPLAINABILITY API

7. Global Explainability

Explains model behavior overall.

Method	Endpoint

V1 CONTRACTS

GET	/api/v1/ml/models/{id}/explain/global/

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8. Local Explainability

Explains **one forecast instance**.

Method	Endpoint
GET	/api/v1/ml/models/{id}/explain/local/{forecast_id}/

Uses:

- SHAP
- Feature importance
- Partial dependence

E. MODEL VERSIONING & REGISTRY API

9. Model Registry

Method	Endpoint
GET	/api/v1/ml/models/
GET	/api/v1/ml/models/{id}/
PUT	/api/v1/ml/models/{id}/
DELETE	/api/v1/ml/models/{id}/

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10. Model Promotion (Controlled)

Role	Endpoint
Data Scientist	POST /api/v1/ml/models/{id}/request-promotion/
SuperAdmin	POST /api/v1/admin/models/{id}/approve/
SuperAdmin	POST /api/v1/admin/models/{id}/reject/

F. DATA DRIFT & MODEL HEALTH API

11. Drift Detection

Method	Endpoint
GET	/api/v1/ml/drift/data/
GET	/api/v1/ml/drift/model/

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12. Retraining Triggers

V1 CONTRACTS

Method	Endpoint
POST	/api/v1/ml/retrain/schedule/
GET	/api/v1/ml/retrain/status/

G. TRAINING AUDIT & REPRODUCIBILITY

13. Training Lineage (Read-only)

Question	Endpoint
Dataset used	/api/v1/audit/ml/dataset/{model_id}/
Feature set	/api/v1/audit/ml/features/{model_id}/
Training config	/api/v1/audit/ml/config/{model_id}/
Code version	/api/v1/audit/ml/code/{model_id}/

MUST HAVE APIS

A. DATA VERSION FREEZING / SNAPSHOT API (CRITICAL)

26. DATA SNAPSHOT & FREEZE API

Access: SUPER_ADMIN, DATA_SCIENTIST (read)

Method	Endpoint
POST	/api/v1/data/snapshots/freeze/
GET	/api/v1/data/snapshots/
GET	/api/v1/data/snapshots/{id}/
POST	/api/v1/data/snapshots/{id}/lock/

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B. FORECAST SCHEDULING & ORCHESTRATION API (VERY IMPORTANT)

27. FORECAST ORCHESTRATION API

Access: SUPER_ADMIN, ECONOMIC_ANALYST

Method	Endpoint
POST	/api/v1/forecasts/schedule/
GET	/api/v1/forecasts/schedules/
PUT	/api/v1/forecasts/schedules/{id}/pause/

V1 CONTRACTS

PUT	/api/v1/forecasts/schedules/{id}/resume/

C. FORECAST APPROVAL & RELEASE API (GOVERNANCE)

28. FORECAST APPROVAL & RELEASE API

Access:

- Analyst → submit
- SuperAdmin / Executive → approve

Method	**Endpoint**
POST	/api/v1/forecasts/{id}/submit/
POST	/api/v1/admin/forecasts/{id}/approve/
POST	/api/v1/admin/forecasts/{id}/reject/
GET	/api/v1/forecasts/{id}/status/

D. ACTUALS vs FORECAST FEEDBACK API

Forecasting systems **must learn from reality**.

ingest sales and **close the loop**.

29. FORECAST FEEDBACK & ACCURACY API

Access: DATA_SCIENTIST, ANALYST

Method	**Endpoint**
POST	/api/v1/forecasts/{id}/actuals/
GET	/api/v1/forecasts/{id}/accuracy/
GET	/api/v1/forecasts/accuracy/trends/

E. ASSUMPTIONS & NOTES API (SURPRISINGLY IMPORTANT)

assumptions shld not be implicit.

30. FORECAST ASSUMPTIONS & NOTES API

Access: ECONOMIC_ANALYST

Method	**Endpoint**
POST	/api/v1/forecasts/{id}/assumptions/
GET	/api/v1/forecasts/{id}/assumptions/
PUT	/api/v1/forecasts/{id}/assumptions/{aid}/

V1 CONTRACTS

F. ALERTS & THRESHOLDS API (OPERATIONAL SAFETY)

Right now dashboards are passive.

31. ALERTS & THRESHOLD API

Access: SUPER_ADMIN, ANALYST

Method	Endpoint
POST	/api/v1/alerts/
GET	/api/v1/alerts/
POST	/api/v1/alerts/{id}/acknowledge/

Updating policy indicators

Resource GET POST PUT DELETE

VAT Rate /api/v1/vat/ /api/v1/vat/ /api/v1/vat/{id}/ /api/v1/vat/{id}/

Corporate Tax /api/v1/corporate-tax/ /api/v1/corporate-tax/ /api/v1/corporate-tax/{id}/ /api/v1/corporate-tax/{id}/

Import Duty /api/v1/import-duty/ /api/v1/import-duty/ /api/v1/import-duty/{id}/ /api/v1/import-duty/{id}/

Fuel Levy /api/v1/fuel-levy/ /api/v1/fuel-levy/ /api/v1/fuel-levy/{id}/ /api/v1/fuel-levy/{id}/

Subsidy Rates /api/v1/subsidy/ /api/v1/subsidy/ /api/v1/subsidy/{id}/ /api/v1/subsidy/{id}/

Sector Tax Incentives /api/v1/tax-incentives/ /api/v1/tax-incentives/ /api/v1/tax-incentives/{id}/ /api/v1/tax-incentives/{id}/

V1 CONTRACTS